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Experiment : 8

Aim:

To set up a cryptocurrency wallet using MetaMask and connect it to a decentralized application (DApp) to execute and verify blockchain transactions.

Theory:

MetaMask is a digital cryptocurrency wallet that enables users to access blockchain networks and interact with decentralized applications. It manages blockchain accounts and allows users to approve or reject transactions requested by DApps.

A Decentralized Application (DApp) is an application that interacts with smart contracts deployed on a blockchain. Its frontend connects to the blockchain through a Web3 provider, such as MetaMask, which serves as a link between the user's browser and the blockchain network.

In this experiment, the ZeroTrace DApp was used to demonstrate the integration of MetaMask with a decentralized application. The DApp uses Web3 technologies to connect to the user's wallet and communicate with deployed smart contracts. After connecting the wallet, a blockchain transaction was initiated through the DApp and approved by the user in MetaMask. The transaction was then verified using its unique transaction hash on a blockchain explorer.

The basic interaction flow is represented as:

User → ZeroTrace DApp → MetaMask Wallet → Smart Contract → Blockchain Network → Transaction Confirmation

MetaMask provides features such as account management, network selection, transaction authorization, and digital signing. Each submitted blockchain transaction receives a unique transaction hash, which can be used to check its status and view its details on a blockchain explorer.

ZeroTrace DApp

ZeroTrace

Features How It Works Architecture

Privacy-Preserving Quadratic Funding

Fair Funding. Private Voting. Verified Humans.

A decentralized platform that enables communities to fairly allocate treasury funds while maintaining complete voter anonymity and preventing sybil attacks.

Get Started → GitHub

ZeroTrace

← Home

Connect Your Wallet

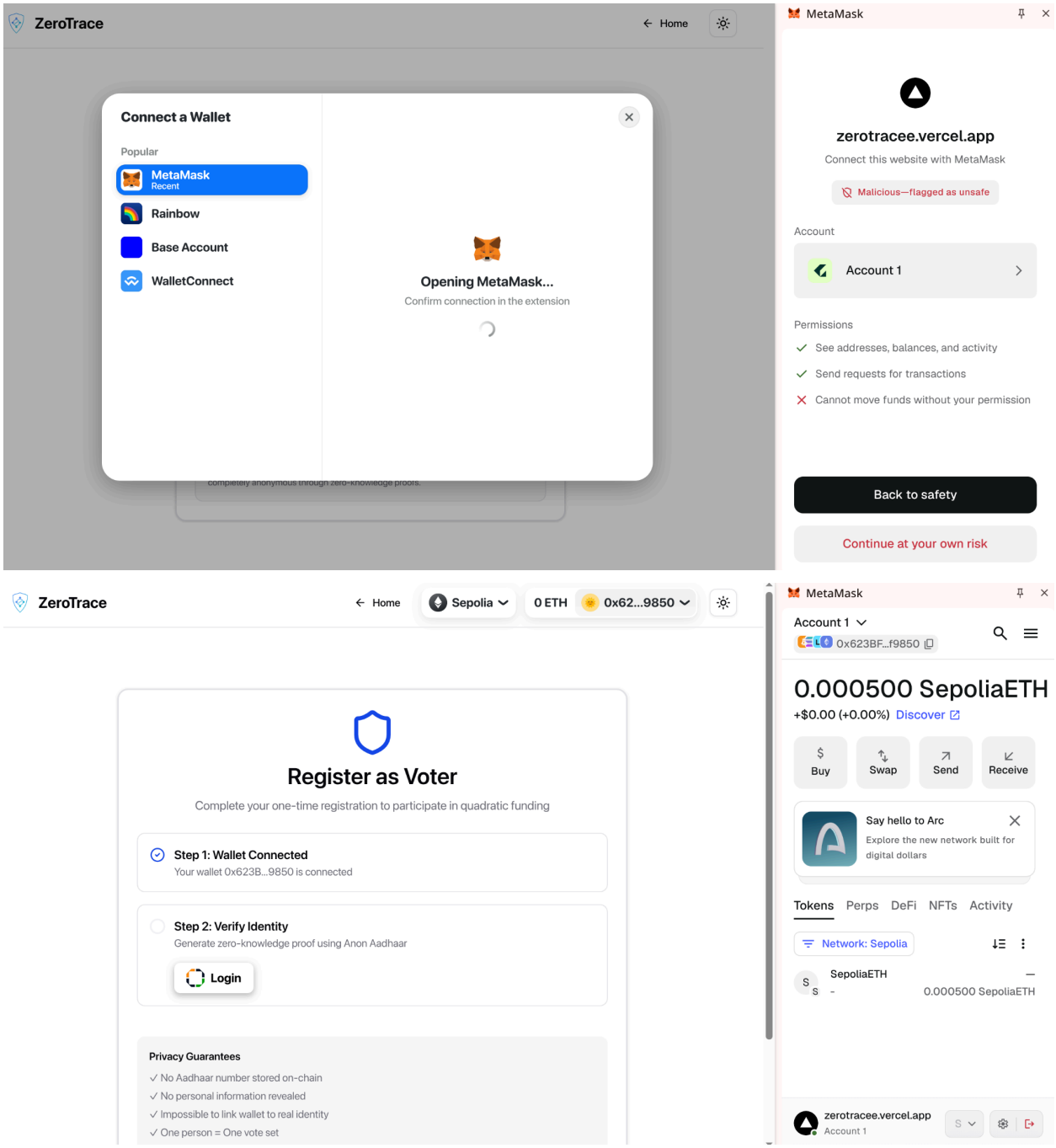
Connect your wallet to access the ZeroTrace platform

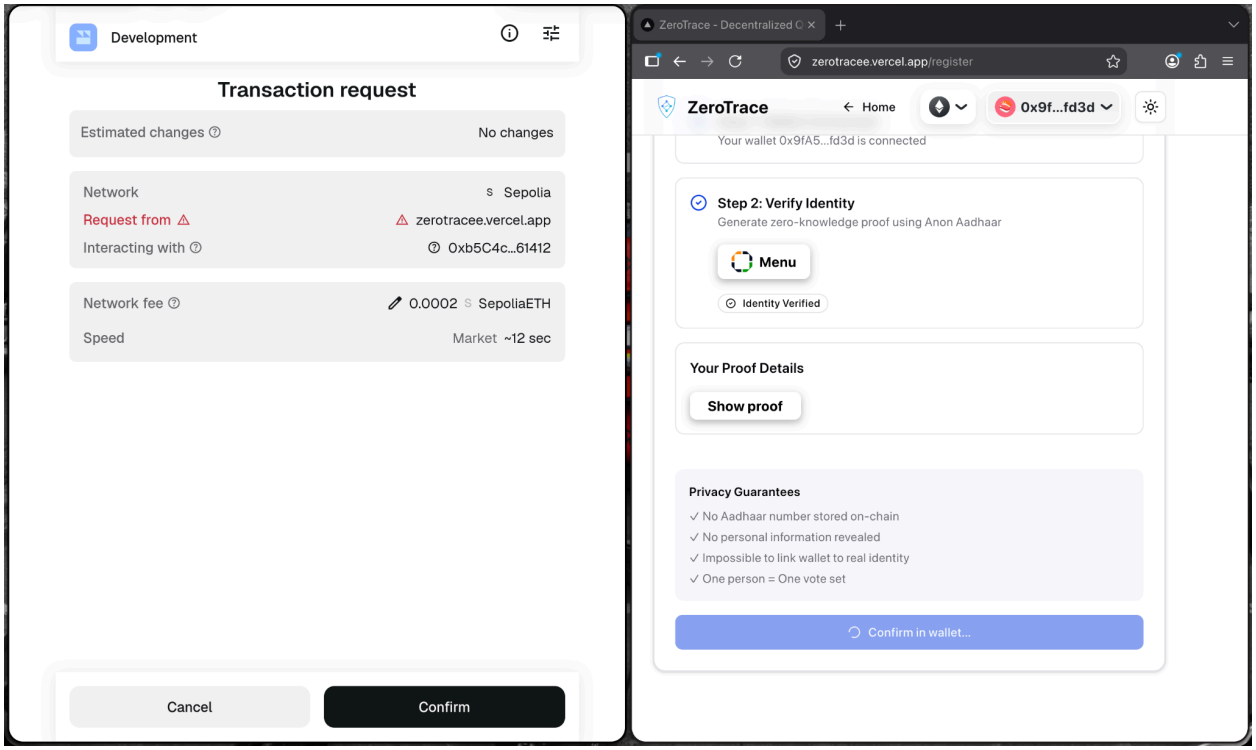
Connect Wallet

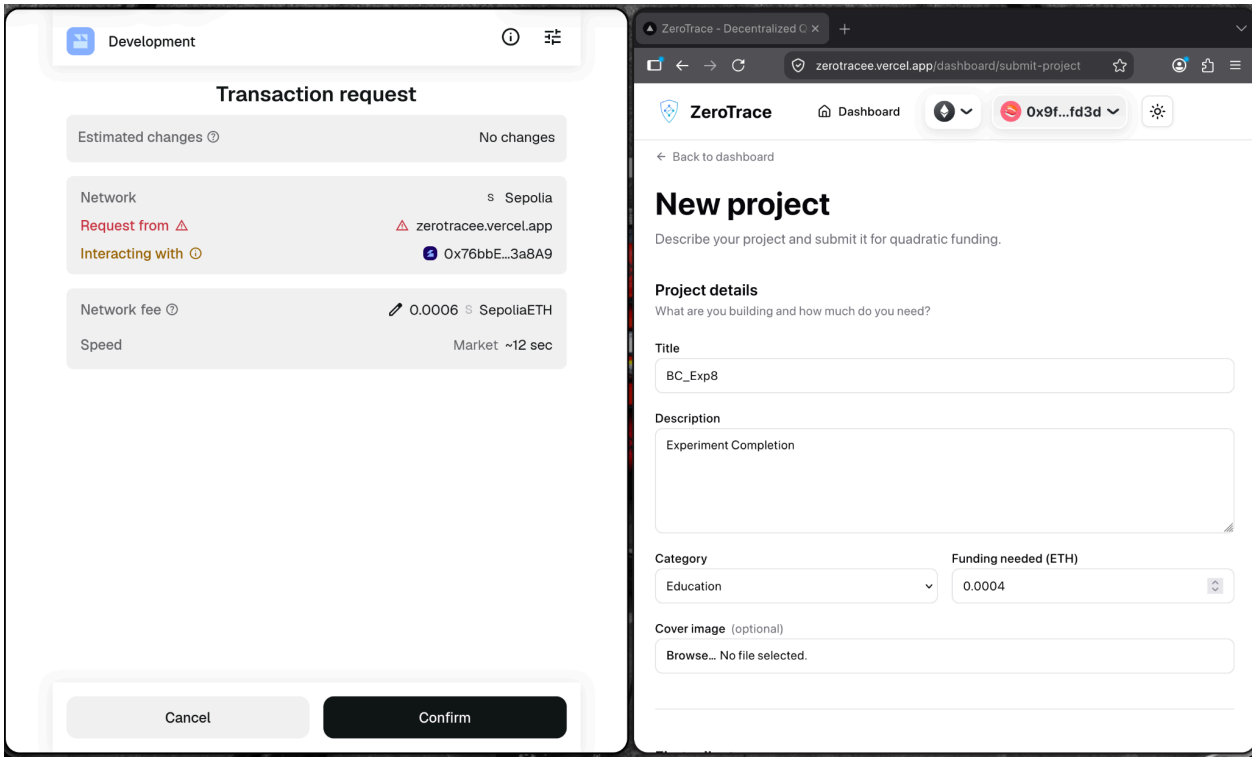
Why connect your wallet?

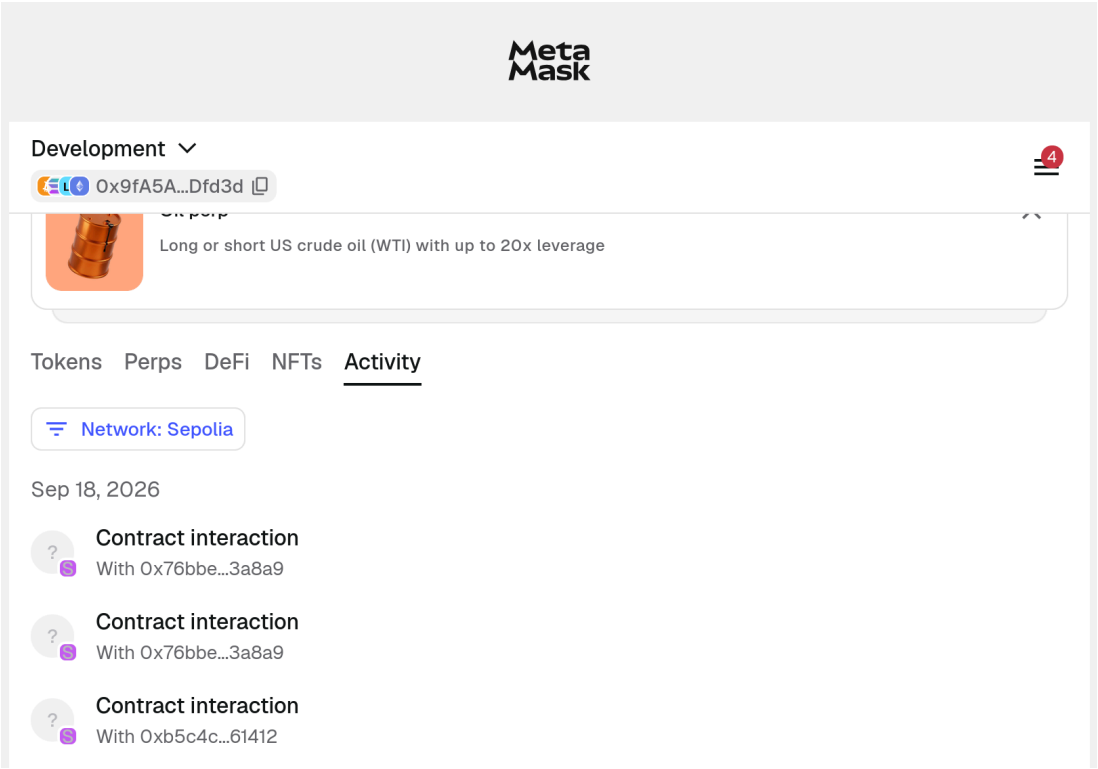
- Register as a verified voter with Anon Aadhaar
- Participate in quadratic funding rounds
- Vote anonymously on project proposals
- Submit your own projects for funding

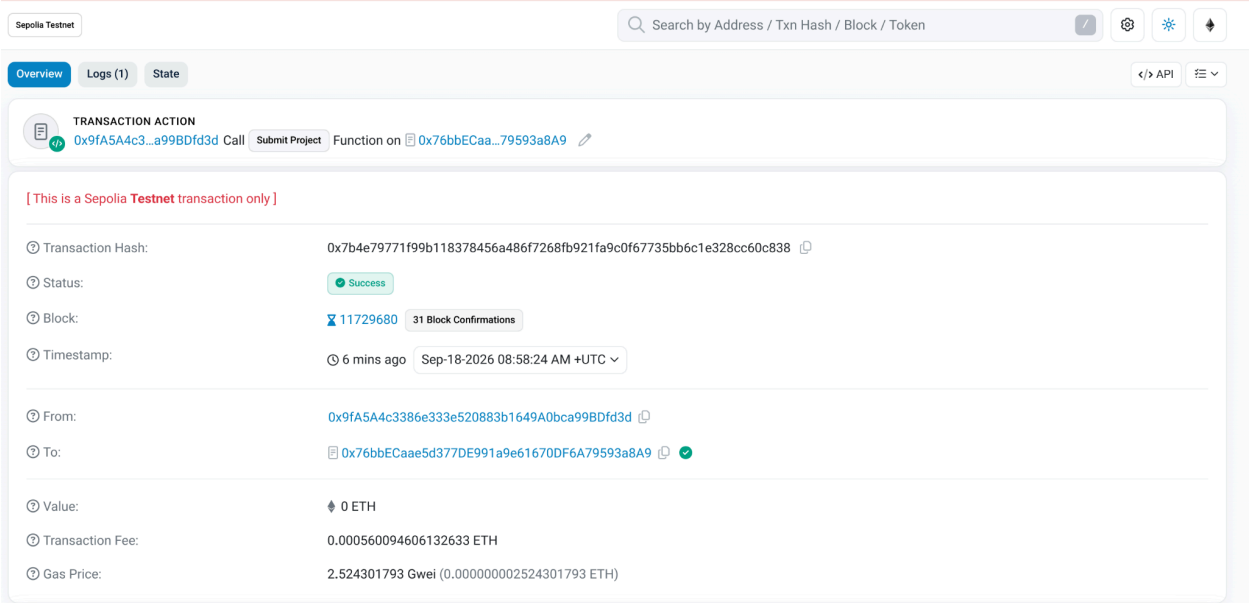
Privacy Notice: Your wallet address is used only for authentication. All votes are completely anonymous through zero-knowledge proofs.


ZeroTrace
Home
Connect a Wallet
Popular
MetaMask
Recent
Rainbow
Base Account
WalletConnect
Opening MetaMask...
Confirm connection in the extension
MetaMask
zerotracee.vercel.app
Connect this website with MetaMask
Malicious—flagged as unsafe
Account
Account 1
Permissions
See addresses, balances, and activity
Send requests for transactions
Cannot move funds without your permission
Back to safety
Continue at your own risk
ZeroTrace
Home
Sepolia
0 ETH
0x62...9850
Register as Voter
Complete your one-time registration to participate in quadratic funding
Step 1: Wallet Connected
Your wallet 0x623B...9850 is connected
Step 2: Verify Identity
Generate zero-knowledge proof using Anon Aadhaar
Login
Privacy Guarantees
✓ No Aadhaar number stored on-chain
✓ No personal information revealed
✓ Impossible to link wallet to real identity
✓ One person = One vote set
MetaMask
Account 1
0x623BF...f9850
0.000500 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Say hello to Arc
Explore the new network built for digital dollars
Tokens
Perps
DeFi
NFTs
Activity
Network: Sepolia
SepoliaETH
0.000500 SepoliaETH
zerotracee.vercel.app
Account 1


Development
Transaction request
Estimated changes
No changes
Network
Sepolia
Request from
zerotracee.vercel.app
Interacting with
0xb5C4c...61412
Network fee
0.0002 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm
ZeroTrace - Decentralized
zerotracee.vercel.app/register
ZeroTrace
Home
0x9f...fd3d
Your wallet 0x9fA5...fd3d is connected
Step 2: Verify Identity
Generate zero-knowledge proof using Anon Aadhaar
Menu
Identity Verified
Your Proof Details
Show proof
Privacy Guarantees
✓ No Aadhaar number stored on-chain
✓ No personal information revealed
✓ Impossible to link wallet to real identity
✓ One person = One vote set
Confirm in wallet...


Development
Transaction request
Estimated changes
No changes
Network
Sepolia
Request from
zerotracee.vercel.app
Interacting with
0x76bbE...3a8A9
Network fee
0.0006 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm
ZeroTrace - Decentralized
zerotracee.vercel.app/dashboard/submit-project
ZeroTrace
Dashboard
0x9f...fd3d
Back to dashboard
New project
Describe your project and submit it for quadratic funding.
Project details
What are you building and how much do you need?
Title
BC_Exp8
Description
Experiment Completion
Category
Education
Funding needed (ETH)
0.0004
Cover image (optional)
Browse... No file selected.





Conclusion:

Thus, the MetaMask cryptocurrency wallet was successfully configured and integrated with the ZeroTrace DApp. The DApp was able to connect to the MetaMask wallet and initiate a blockchain transaction, which was approved by the user through MetaMask. The transaction was successfully recorded and verified using its transaction ID on the blockchain explorer. This experiment demonstrated the interaction between a cryptocurrency wallet, a decentralized application, smart contracts, and a blockchain network.